1. Briefly define a queue data structure.

A queue is a first-in first-out (FIFO) data structure, where new items are added to the back and items are popped from the front.

What evidence did you find that a queue container was operating in the load balancer?

Finished at 164 E pid 5/20 process time was 146 processed request from 10.34.188.121 to 58.59.194.39
Finished at 205 A pid 1/22 process time was 191 processed request from 237.34.169.94 to 167.101.86.233
Finished at 246 B pid 2/25 process time was 232 processed request from 24.127.73.237 to 222.26.236.209
Finished at 305 A pid 7/28 process time was 99 processed request from 252.14.0.93 to 97.129.102.93
Finished at 327 C pid 3/30 process time was 312 processed request from 108.203.29.136 to 113.114.6.216
Finished at 378 D pid 4/34 process time was 363 processed request from 83.129.9.252 to 110.43.165.204
Finished at 419 E pid 6/37 process time was 251 processed request from 221.19.235.109 to 127.182.138.8

Looking at the number following pid, we can see that lower pid numbers tend to be processed first. Since the load is split up between 5 servers, however, it isn't exactly in-order.

2. Briefly define a stack data structure.

A stack is a last-in last-out (LILO) data structure, where new items are placed in the back and items are popped from the back as well.

What evidence did you find that a stack container was operating in the load balancer?

Finished at 497 C pid 8/36 process time was 482 processed request from 138.62.237.143 to 33.146.29.130
Finished at 631 B pid 26/45 process time was 384 processed request from 103.19.42.212 to 211.17.255.70
Finished at 677 C pid 48/48 process time was 176 processed request from 218.164.29.69 to 228.103.76.156
Finished at 689 E pid 28/47 process time was 427 processed request from 65.180.164.84 to 222.120.39.239
Finished at 729 E pid 61/50 process time was 35 processed request from 211.157.100.244 to 137.69.63.27

Again looking at the number following pid, we can see that the new requests are being processed before the old ones. They were the last in, so they went out first.

3. Briefly define a priority queue data structure.

A priority queue processes items in order of their assigned priority. This is often achieved via minimum and maximum heaps, where items with the lowest or highest number get processed first respectively.

What evidence did you find that a Max priority queue based on process time was operating in the load balancer?

Finished at 970 A pid 39/88 process time was 463 processed request from 243.120.159.130 to 159.145.176.179
Finished at 1359 E pid 81/118 process time was 486 processed request from 17.215.116.103 to 24.10.255.151
Finished at 1372 C pid 89/118 process time was 467 processed request from 84.8.240.232 to 56.188.27.51
Finished at 1383 D pid 94/118 process time was 492 processed request from 15.4.137.35 to 47.160.29.172
Finished at 1406 B pid 75/120 process time was 462 processed request from 193.231.4.226 to 127.130.253.122
Finished at 1430 A pid 83/123 process time was 459 processed request from 57.19.99.44 to 36.116.4.152

Looking at the numbers after “process time was”, we can see that process times were generally in the upper 400s, which is very notable because we pseudo-randomly generated numbers between 0 and 499.

What evidence did you find that a Min priority queue based on process time was operating in the load balancer?

Finished at 626 B pid 41/27 process time was 84 processed request from 186.176.152.201 to 182.117.20.83
Finished at 643 D pid 49/26 process time was 37 processed request from 108.63.156.180 to 185.199.16.162
Finished at 680 A pid 45/29 process time was 93 processed request from 133.137.37.169 to 143.47.36.175
Finished at 705 A pid 54/28 process time was 22 processed request from 79.128.225.36 to 168.48.103.15
Finished at 727 C pid 42/31 process time was 118 processed request from 38.149.119.182 to 55.62.255.208
Finished at 757 C pid 58/31 process time was 29 processed request from 166.231.211.207 to 73.9.231.101

The numbers generated here for process time are much lower on average compared to the max priority queue.

What evidence did you find that a Max priority queue based on process ID was operating in the load balancer?

Finished at 1731 B pid 121/126 process time was 495 processed request from 17.84.165.92 to 39.237.138.233
Finished at 1739 E pid 138/126 process time was 327 processed request from 157.160.108.231 to 57.244.238.201
Finished at 1768 D pid 147/130 process time was 262 processed request from 222.104.240.33 to 209.11.3.64
Finished at 1812 C pid 137/129 process time was 399 processed request from 22.152.60.204 to 136.79.128.250
Finished at 1975 A pid 163/147 process time was 261 processed request from 118.18.177.135 to 39.175.3.135
Finished at 2087 C pid 169/156 process time was 273 processed request from 255.182.135.162 to 54.57.106.205

The pids are steadily increasing, and higher pids are getting processed first.